

ShopSphere

S24 - Performance, Full Regression and Production Release

Backend + DB

Platform-Specific Sprint Planning Document

This document intentionally defines only the behavior and delivery scope of the selected platform. It does not mix implementation details from the other platforms.

1. Sprint Definition

Sprint: S24 - Performance, Full Regression and Production Release

Platform: Backend + DB

Application: ShopSphere

Objective: Implement the backend and database boundary for Performance, Full Regression and Production Release, including business rules, persistence and API behavior.

This sprint is complete only when the stated use cases, acceptance criteria and tests pass for this platform.

2. Business Use Cases

ID	Use Case	Actor	Precondition	Main Flow	Expected Result
UC-01	Release production	Release owner	All required gates passed	Deploy approved builds/configuration, run smoke tests	Production is serving the expected version and monitoring is healthy.
UC-02	Recover from failure	Release/ops	Rollback criteria met	Execute rollback/recovery procedure	Service is restored without data corruption.

3. Detailed Implementation Plan

1. Run load/performance tests on critical APIs.
2. Tune indexes, connection pooling, Redis caching and slow queries.
3. Verify backups, restore, health checks, alerts and logs.
4. Run full API/integration/regression suite.
5. Validate production migration and rollback plan.

4. Data / State / Contract Scope

- production configuration and monitoring metadata

API / Integration Contract

- Feature APIs for this sprint, versioned under /api/v1.
- Every endpoint must define authentication, authorization, validation, error codes and request correlation.

5. Rules and Expected Behavior

- The platform must rely on the backend as the source of truth for business rules.
- Do not expose stack traces, secrets or internal implementation details to end users.
- Handle loading, empty, validation and error states explicitly.
- Requests that can be retried must be safe from duplicate side effects where the feature requires idempotency.
- Ownership and authorization must be checked server-side for protected data.

6. Acceptance Criteria

- Data model supports the required business relationships and constraints.
- API enforces authentication, authorization, validation and ownership where applicable.
- Business rules are server-authoritative.
- Transactions/idempotency/concurrency controls are used where the feature requires them.
- Controlled error codes and request correlation are implemented.

7. Test Cases

TC	Scenario	Expected Result
TC-S24-01	Load spike	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S24-02	Database failure	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S24-03	Redis failure	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S24-04	Payment outage	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S24-05	Shipping outage	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S24-06	Notification outage	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S24-07	Rollback migration	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S24-08	Expired secret	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S24-09	Production API misconfiguration	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S24-10	Mobile release build failure	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S24-11	Admin export failure	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.

8. Sprint Exit Checklist

- Implementation completed only for this platform.
- All primary and negative use cases pass.
- API contracts are implemented/consumed as defined.
- Validation and authorization tests pass.
- Loading/empty/error handling is verified where applicable.
- No critical or blocker defect remains.
- Automated tests pass in CI.
- Documentation and contract changes are committed.

9. Out of Scope for This Document

- Implementation behavior of Customer Portal. That behavior belongs in the separate S24 document for that platform.
- Implementation behavior of Admin Portal. That behavior belongs in the separate S24 document for that platform.